

Stat380 Final Report Group 3

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1. Introduction

Our project focuses on the United States' market. We know that financial markets are sensitive to not only the contemporary economy state, but also how new information compares to what investors expected. The macroeconomic announcements like the Consumer Price Index (CPI), the unemployment rate, and Federal Reserve interest rate decisions are among the most closely monitored. These events are important because they can change expectations about inflation, economic growth, and future interest rates. When these pivotal changes are apparent, investors may adjust their portfolios, which can cause the stock market volatility to go either up or down.

In this project, we want to study the effects of these macroeconomic announcements towards the short-term volatility in the U.S stock market. We would be using the SPDR S&P 500 ETF (SPY) as our overall representation of the stock market. Instead of just looking at the average returns, we focus on the realized volatility around the macroeconomic announcement dates.

We compute daily log returns for SPY and use the absolute value of these returns as a daily volatility proxy. We then examine how this volatility behaves before and after CPI, unemployment, and federal funds rate announcements.

Our main research question is: "How do month-to-month changes in key U.S. macroeconomic indicators (CPI, unemployment, and the Federal funds rate) relate to the short-term changes in SPY volatility around announcement dates?"

As a group, we were motivated to work on this topic because many of us are interested in learning about investing and follow macroeconomic news. We regularly see headlines in articles and television about CPI, unemployment, and Fed decisions "moving the market", but we want to closely see the effect of each of these events carefully by using data to prove its importance to the volatility of the market. This topic also lets us apply several methods we learned from STAT 380. For example, we can use regression and expand into time series models to measure these effects. In this way, the project is both personally interesting to us and a good practice to use what we learn in class to real-world situations.

2. Data Description –

We have four datasets that we are looking at: CPI, Federal Fund Rate, Unemployment, and SPY.

For the CPI, Federal Fund Rate, and Unemployment datasets, they each follow this data structure.

	Date	Actual	Previous
1	Oct 24, 2025	NA	0.3
2	Aug 12, 2025	0.3	0.2

The variables are Date, Actual, and Previous.

Date: The date in which the CPI, Federal Fund Rate, or Unemployment is released.

Notice that the Federal Fund Rate is only released 8 times per years because is based on the Federal Open Market Committee.

Actual: The new value

Notice that for CPI the actual value is the previous months CPI. For example Oct 24, 2025 releases the CPI for September NOT October. Therefore, when interpreting, understand we are looking at how the previous months consumer spending affects the market volatility

Previous: Previous time periods amount

In the SPY dataset, the variables are the following:

	X	SPY.Open	SPY.High	SPY.Low	SPY.Close	SPY.Volume	SPY.Adjusted
1	2000-01-03	148.2500	148.2500	143.8750	145.4375	8164300	91.88779
2	2000-01-04	143.5312	144.0625	139.6406	139.7500	8089800	88.29438

The variables are X, SPY.Open, SPY.High, SPY.Low, SPY.Close, SPY.Volume, and SPY.Adjusted.

X: Date

Since we are looking at market volatility, we will need to compute the daily log returns.

Daily log returns = $\ln(\text{SPY.Adjusted day} / \text{SPY.Adjusted previous day})$.

We need to take the absolute value for the daily volatility proxy. Once we compute this, we create a new column called `spy_vol`

```
[1] "X"           "SPY.Open"    "SPY.High"    "SPY.Low"     "SPY.Close"
[6] "SPY.Volume"  "SPY.Adjusted" "ret"         "spy_vol"
```

Check for NA values

- CPI has two missing values (actual Oct 24, 2025 and previous Nov 1, 1999).
- Federal Reserve Rate has two missing values (actual Dec 10, 2025 and previous Jun 30, 1999)
- Unemployment has two missing values (actual Nov 20, 2025 and previous May 1, 1999)
- SPY has two missing value (ret and spy_vol Jan 3, 2000)

Since each dataset has only two missing values in 1999 and 2025, we will drop these rows.

Next, we will merge all the data together into a dataset called `macro_all` in conducting EDA.

In order to do this, we will change the date formatting to year-month-day, add a new difference column (Actual - Previous), and create a name indicator column (CPI, Fed, Unemployment).

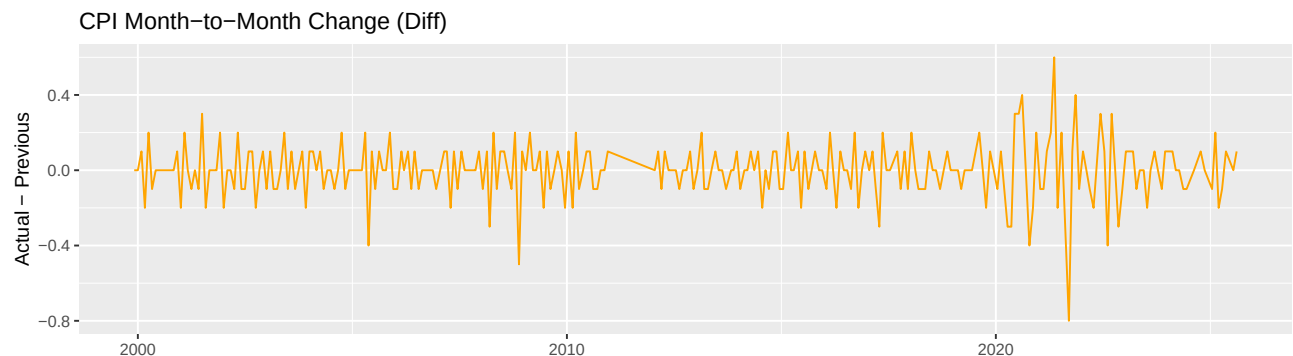
	Date	Actual	Previous	Diff	Indicator
1	2025-08-12	0.3	0.2	0.1	CPI

Summary Statistics

A tibble: 3 x 8

Indicator	n	mean_actual	sd_actual	mean_diff	sd_diff	min_diff	max_diff
<chr>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 CPI	282	0.197	0.147	3.55e- 4	0.145	-0.8	0.6
2 Fed	173	2.03	1.94	-5.78e- 3	0.276	-1	0.75
3 Unemployment	304	5.49	1.86	5.84e-18	0.651	-2.20	10.3

CPI Month-to-Month Change (Diff)



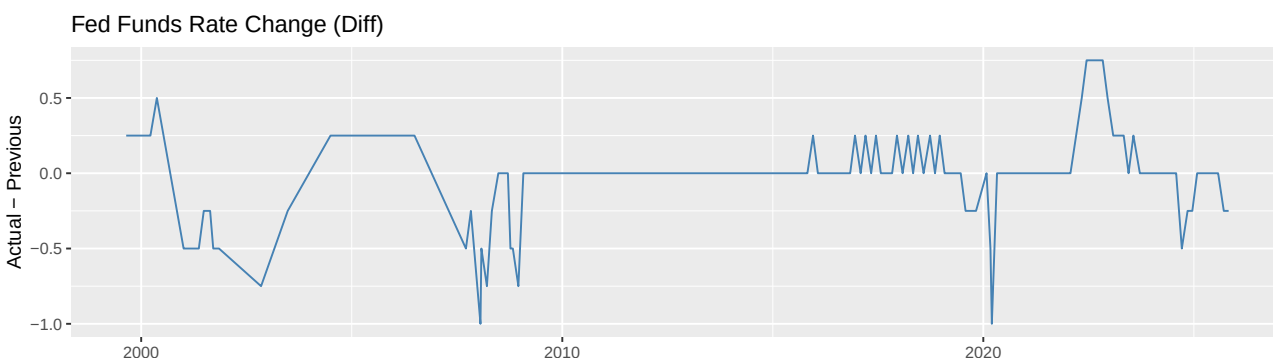
What the plot shows

This plot displays the month-to-month change in the Core CPI (Actual - Previous) from 1999 to 2025. Each point shows how much inflation changed from the prior month.

Interpretation

CPI changes are very small and stable, almost always between -0.2 and $+0.2$. This indicates that inflation moves gradually, with very few large shocks. Notable drops occur during financial crises (2008) and the COVID period (2020), but otherwise CPI is smooth and predictable. Because CPI changes are so modest, their impact on market volatility is likely weaker and more subtle compared to the Fed announcements.

Fed Funds Rate Change (Diff)



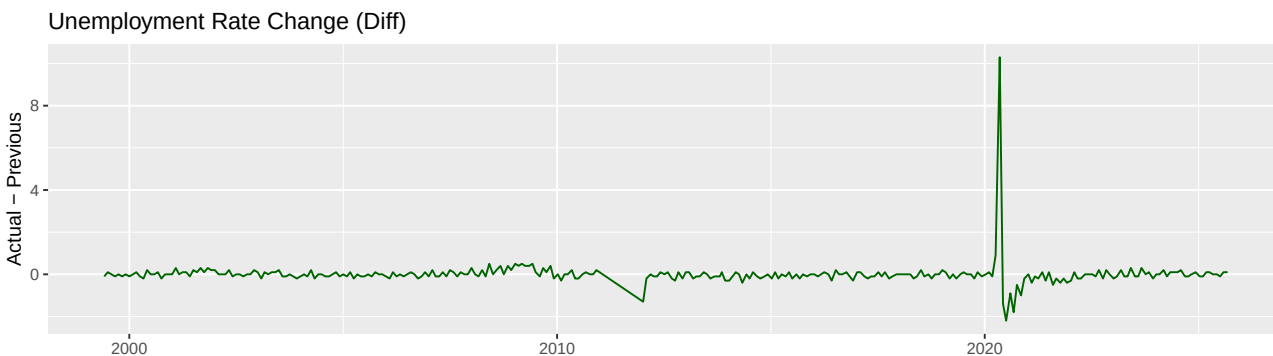
What the plot shows

This plot shows month-to-month changes in the Federal Funds Rate determined by the Federal Reserve, represented as Actual — Previous across Fed meeting dates.

Interpretation

Fed rate changes are infrequent but large. Long stretches of zero change are followed by sudden rate hikes or cuts, especially around recessionary periods (early 2000s, 2008–2009) and during the COVID crash (2020). These “jump” behaviors indicate that Fed decisions introduce significant shocks into financial markets. This makes the Fed Funds Rate one of the most influential macro indicators for market volatility.

Unemployment Rate Change (Diff)



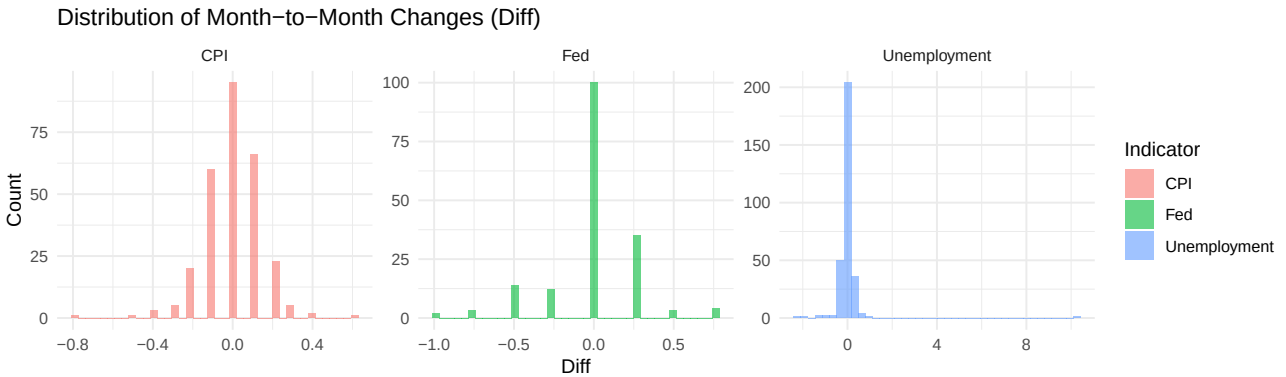
What the plot shows

This line plot shows the month-to-month change in the U.S. unemployment rate from 1999 to 2025.

Interpretation

Unemployment changes are extremely stable, typically moving only ± 0.1 month-to-month. The single massive spike in 2020 reflects the COVID employment crisis, where unemployment rose sharply in one month. Outside of this event, unemployment tends to change very slowly. Because unemployment rarely jumps, it likely has the weakest short-term effect on SPY volatility except during extreme economic disruptions.

Distribution of Month-to-Month Changes (Histograms)



What the plot shows

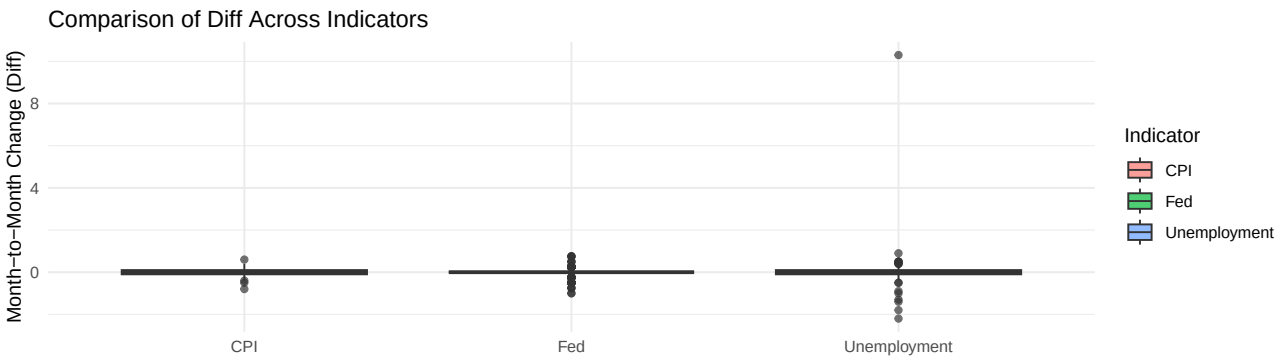
Three histograms (CPI, Fed, Unemployment) visualize the distribution of their Diff values (Actual – Previous).

Interpretation

- **CPI:** Very tight symmetric distribution around zero, indicating consistent and predictable monthly changes.
- **Fed:** Clustered at zero, with occasional strong positive or negative changes, showing the Fed creates discrete shocks.
- **Unemployment:** Extremely concentrated at zero except for a few rare large jumps, especially the COVID spike.

These distributions highlight that macro indicators have very different statistical behaviors, supporting treating them as separate predictors in the model.

Boxplot Comparison Across Indicators



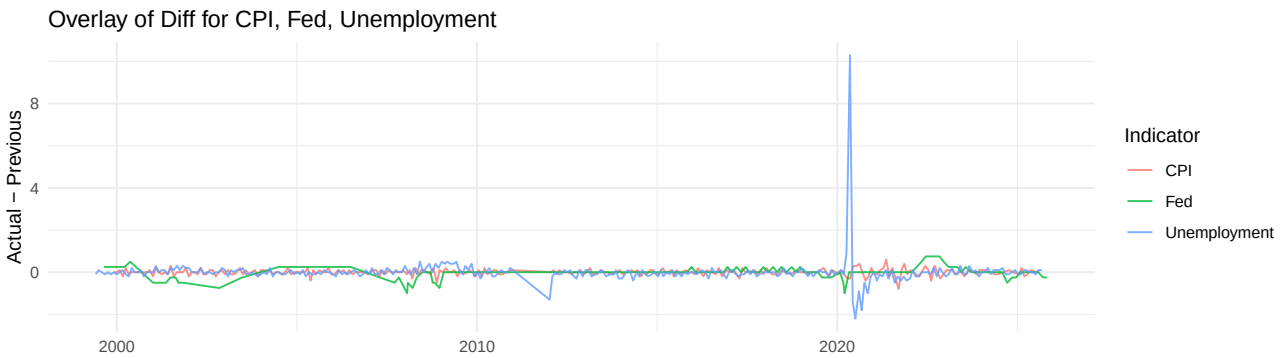
What the plot shows

This boxplot compares the variance and outliers in Diff across CPI, Fed Funds Rate, and Unemployment Rate.

Interpretation

CPI has the smallest range with minimal outliers, reflecting stable inflation. The Fed series shows larger spread and multiple outliers, indicating significant rate adjustments. Unemployment has one extreme outlier (COVID spike) but otherwise behaves similarly to CPI. These differences confirm that each macro indicator has a distinct volatility pattern, which directly affects how markets respond to each announcement.

Overlay of Diff for CPI, Fed, and Unemployment



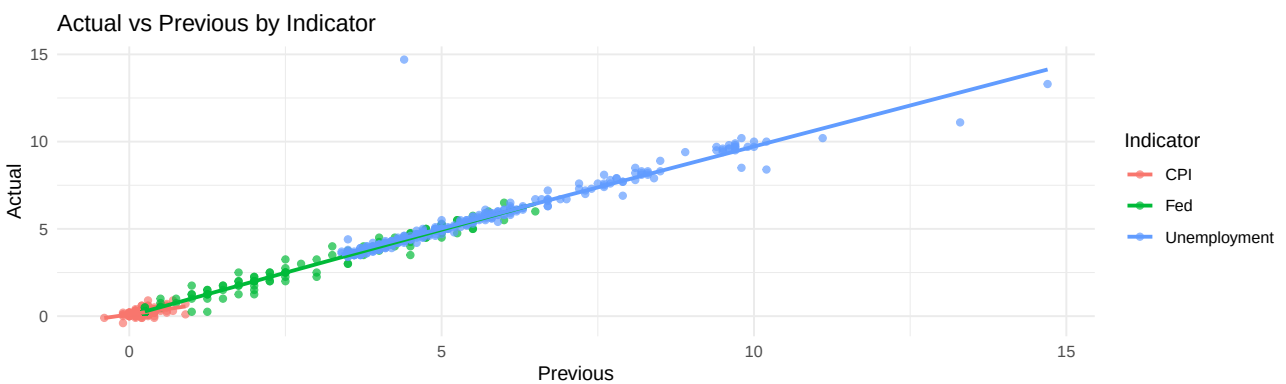
What the plot shows

All three Diff series are plotted together to show how they behave over time and relative to each other.

Interpretation

All three indicators remain close to zero most of the time, but the Fed series exhibits sudden jumps, and unemployment has one massive spike in 2020. CPI fluctuates within a very narrow band. The overlay makes the structural contrast clear: Fed decisions cause jump shocks; CPI is smooth and stable; unemployment is stable except in crisis conditions. This difference justifies comparing their individual effects on SPY volatility.

Actual vs Previous Scatterplot



What the plot shows

This scatterplot shows the relationship between the current month's value (Actual) and the previous month's value (Previous) for CPI, Fed, and Unemployment.

Interpretation

All three indicators lie almost perfectly along a 45-degree line, meaning $\text{Actual} \approx \text{Previous}$ for most months. This strong autocorrelation confirms that these macroeconomic indicators move slowly over time, with rare large jumps. Because of this, the month-to-month difference (Diff) captures essentially all meaningful new information, validating its use as the key predictor in the volatility model.

3. Methodology –

Regression Analysis:

Our goal is to create a multilogistical regression model where the response variable is SPY volatility and the independent variables are the CPI, Federal Fund Rate, and the Unemployment monthly difference (Actual - Predicted).

We want to specifically look at the daily SPY volatility between 2008 and 2015 because this period reflects the global financial crisis. Therefore, looking at this special time period gives us key insight into market volatility during that period.

When creating this model, we discussed how each variable (Unemployment, Federal Fund Rate, and CPI) all come out at different times throughout the month. One of the challenges in this is having a model that simultaneously

looks at each of these variables. We need all these variables in one model to determine the changes in market volatility when each of these values are released.

In order to track whether or not a variable came out, we used dummy variables. Within each month, Dummy=1 indicating the value is released. The dummy variable stays till one until the beginning of the next month, where all the variables reset to Dummy=0. Knowing which information is publically available enables us to correctly analyze market volatility. We will specially be looking at the market volatility the day after the rates are released.

a) Our first model looks at how much the market volatility changes based on each variable

b) Next, we are going to see how the performance changes whether or not all three variables come out within that month.

In our study, we acknowledge that these are only a few drivers of the SPY. However, for this exploration, our focus is on how much these factors move the market when they are first released

Time-Series Analysis:

For our other statistical model, we utilized a time-series model – which is beyond our class material. The time-series model focuses on the daily SPY volatility between 2008 and 2015. For context, this is an era where there was a global financial crisis and the following recovery years for the country. In this model, we included four datasets, as mentioned from previous sections of this paper: the daily SPY prices from `SPY_data.csv`, and the monthly macroeconomic data from `cpi_cleaned.csv`, `unemployment_cleaned.csv`, and `fed_rate_cleaned.csv`. Inside the SPY data, we did some calculations to figure out the daily log returns using the adjusted closing price, and then defined a less complex volatility representation as the absolute value of the log return for each day. This allows us to have a time series of daily volatility called `spy_vol`. For the other variables derived from the macroeconomic data, we created “surprise” metrics which is the difference between current release and the previous value (Actual - Previous) for each of the independent variables: CPI, unemployment, and the Federal funds rate. Later, we merged these surprises onto the daily SPY data by date. If there were days that there were no macro releases, we set these surprise values to zero so that the series remained daily and aligned with the SPY volatility data.

With the construction of the merged dataset, we reorganized it by data and split the observations into a training and test set. The training set contained 1,409 observations (70% of the data) and the test set contained the remaining 605 observations (30% of the data). All of the models that we included in this part were fitted on the training data and evaluated on the test data. To analyze the relationship between macro surprises over time and volatility, we estimated two different time-series models: an ARIMAX model, which extends the ARIMA framework, and a GARCH(1,1) model, which is a tool used for financial econometrics.

The ARIMAX model would treat the daily volatility as the response variable and uses the macro surprises and its own past behavior as the predictors. We modeled daily volatility as a regression with ARIMA errors. The ARIMA would capture autocorrelation and persistence in volatility, with the predictors being CPI, unemployment, and Fed funds rate surprises on the same day. We would use the function, `auto.arima()`, and the response variable being the volatility series; the three surprise variables would be included in “xreg”, without a seasonal component. The function would automatically choose the AR and MA orders based on information criteria. It would evidently choose ARIMA(5,1,4). We decided to use the ARIMAX model because it helps us in understanding whether macro surprises provide extra predictive information for volatility beyond what can be explained by past volatility alone.

GARCH(1,1) was another time-series model we used, which was part of the `rugarch` package. In this model, the today's volatility is dependent on three things: a constant, yesterday's squared shocked, and yesterday's volatility. This is like a normal setup found in finance because it helps with modeling volatility clustering, where periods of high volatility often shows up in consecutive days instead of just a single random spike. In our project, we used a simple GARCH(1,1) model on the daily volatility series and did not include the macro surprises directly in the variance equation because we want to understand how volatility behaves on its own. For both models, the ARIMAX and GARCH models, we generated forecasts of daily volatility on the test data and evaluate how accurate they were using RMSE (root mean squared error) and MAE (mean absolute error), where a small value for both metrics means that the model's predictions are close to the true volatility.

4. **Data Analysis Results** – Present and interpret your main findings. Include relevant figures and tables that support your discussion.

Regression Analysis

a) Linear Regression

```
Call:
lm(formula = spy_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day +
    fed_diff + fed_day + all_three, data = data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.023296	-0.006479	-0.003036	0.003092	0.126809

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.0097476	0.0005098	19.119	< 2e-16 ***
cpi_diff	-0.0023874	0.0034614	-0.690	0.4904
cpi_day	0.0002472	0.0007065	0.350	0.7264
unemp_diff	0.0093754	0.0011755	7.975	2.50e-15 ***
unemp_day	-0.0009799	0.0006794	-1.442	0.1494
fed_diff	-0.0173881	0.0023690	-7.340	3.07e-13 ***
fed_day	0.0019476	0.0014075	1.384	0.1666
all_three	-0.0031645	0.0016151	-1.959	0.0502 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01147 on 2051 degrees of freedom
Multiple R-squared: 0.06054, Adjusted R-squared: 0.05733
F-statistic: 18.88 on 7 and 2051 DF, p-value: < 2.2e-16

R²: 0.06053553

Adjusted R²: 0.05732916

AIC: -12546.47

BIC: -12495.8

RMSE: 0.01144676

MAE: 0.0072255

	term	estimate	std.error	t.value	p.value
cpi_diff	cpi_diff	-0.002387403	0.006565589	-0.3636236	7.161766e-01
unemp_diff	unemp_diff	0.009375424	0.001224310	7.6577235	2.895793e-14
fed_diff	fed_diff	-0.017388149	0.004640057	-3.7474002	1.836154e-04
	pct_change_vol				
cpi_diff		-0.2387403			
unemp_diff		0.9375424			
fed_diff		-1.7388149			

A one-unit CPI difference changes daily SPY volatility by -0.24% (robust SE = 0.0066).

A one-unit Unemployment difference changes daily SPY volatility by 0.94% (robust SE = 0.0012).

A one-unit Fed Funds Rate difference changes daily SPY volatility by -1.74% (robust SE = 0.0046).

On CPI release days, daily SPY volatility is estimated to change by 0.02%.

On Unemployment report days, daily SPY volatility is estimated to change by -0.10%.

On Fed rate decision days, daily SPY volatility is estimated to change by 0.19%.

[1] "On days when CPI, unemployment, and the Fed decision are all released, volatility changes by -0.32%."

b) Logistical Regression

Median of spy_vol: 0.005756937

Label counts:

```

      0      1
1030 1029

```

Label proportions:

```

      0      1
0.5002428 0.4997572

```

Preview of labels:

```

      spy_vol high_vol
1 0.008793180      1
2 0.008793180      1
3 0.000483235      0
4 0.000483235      0
5 0.024811454      1
6 0.024811454      1

```

Call:

```

glm(formula = high_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day +
     fed_diff + fed_day + all_three, family = binomial, data = train_data)

```

Coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.13631    0.09994   1.364  0.17260
cpi_diff      0.19176    0.70899   0.270  0.78680
cpi_day     -0.02803    0.14018  -0.200  0.84150
unemp_diff    1.64023    0.26466   6.198 5.73e-10 ***
unemp_day    -0.20142    0.13405  -1.503  0.13295
fed_diff     -1.55012    0.54883  -2.824  0.00474 **
fed_day      -0.14143    0.27385  -0.516  0.60555
all_three     0.13583    0.31609   0.430  0.66740
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```

Null deviance: 2284.6  on 1647  degrees of freedom
Residual deviance: 2226.1  on 1640  degrees of freedom
AIC: 2242.1

```

Number of Fisher Scoring iterations: 4

R²:

Adjusted R²:

AIC: 2242.064

BIC: 2285.323

RMSE: 0.01144676

MAE: 0.0072255

Confusion Matrix and Statistics

```

      Reference
Prediction  0      1
      0  96  80

```


1 110 125

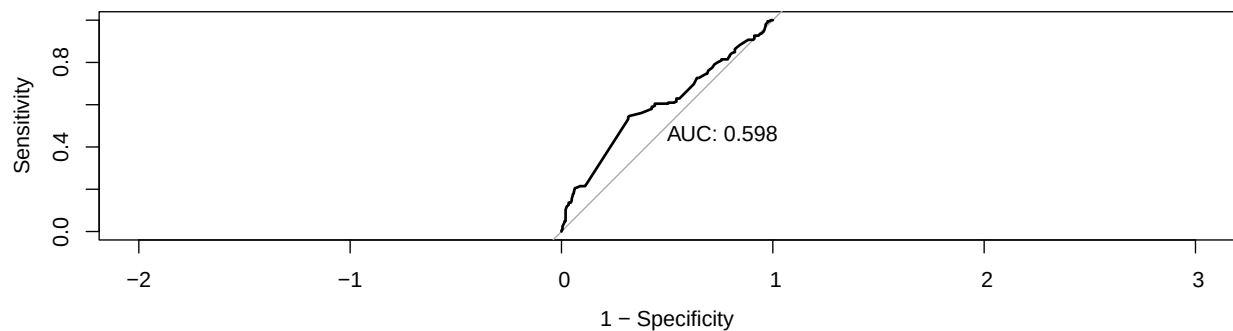
Accuracy : 0.5377
95% CI : (0.4882, 0.5867)
No Information Rate : 0.5012
P-Value [Acc > NIR] : 0.07624

Kappa : 0.0757

McNemar's Test P-Value : 0.03539

Sensitivity : 0.6098
Specificity : 0.4660
Pos Pred Value : 0.5319
Neg Pred Value : 0.5455
Prevalence : 0.4988
Detection Rate : 0.3041
Detection Prevalence : 0.5718
Balanced Accuracy : 0.5379

'Positive' Class : 1



[1] 0.5984845

Time-Series Analysis

TRAIN: 1441 obs TEST: 618 obs

===== ARIMAX Model =====

Series: spy_vol_ts_train

Regression with ARIMA(5,1,1) errors

Coefficients:

	ar1	ar2	ar3	ar4	ar5	ma1	cpi_diff	unemp_diff
	0.0414	-0.1101	-0.0247	0.1420	0.0582	-0.9012	-0.0097	0.0013
s.e.	0.0355	0.0341	0.0343	0.0323	0.0300	0.0229	0.0043	0.0019
	fed_diff							
	0.0019							
s.e.	0.0026							

sigma^2 = 0.0001177: log likelihood = 4474.54

AIC=-8929.08 AICc=-8928.93 BIC=-8876.36

ARIMAX Model Performance

RMSE: 0.00558

MAE : 0.003987

===== GARCH(1,1) Fit =====

```
*-----*
*          GARCH Model Fit          *
*-----*
```

Conditional Variance Dynamics

GARCH Model : sGARCH(1,1)
Mean Model : ARFIMA(0,0,0)
Distribution : norm

Optimal Parameters

 Estimate Std. Error t value Pr(>|t|)
mu 0.007080 0.000206 34.29671 0.00000
omega 0.000001 0.000001 0.96358 0.33526
alpha1 0.110148 0.020687 5.32436 0.00000
beta1 0.884217 0.018356 48.16928 0.00000

Robust Standard Errors:

 Estimate Std. Error t value Pr(>|t|)
mu 0.007080 0.000361 19.61745 0.00000
omega 0.000001 0.000007 0.14353 0.88587
alpha1 0.110148 0.113943 0.96669 0.33370
beta1 0.884217 0.105007 8.42053 0.00000

LogLikelihood : 4788.424

Information Criteria

Akaike -6.6404
Bayes -6.6258
Shibata -6.6404
Hannan-Quinn -6.6350

Weighted Ljung-Box Test on Standardized Residuals

 statistic p-value
Lag[1] 13.58 2.284e-04
Lag[2*(p+q)+(p+q)-1] [2] 38.10 8.031e-11
Lag[4*(p+q)+(p+q)-1] [5] 105.05 0.000e+00
d.o.f=0
H0 : No serial correlation

Weighted Ljung-Box Test on Standardized Squared Residuals

 statistic p-value
Lag[1] 1.418 0.23368
Lag[2*(p+q)+(p+q)-1] [5] 8.497 0.02215
Lag[4*(p+q)+(p+q)-1] [9] 10.666 0.03601
d.o.f=2

Weighted ARCH LM Tests

	Statistic	Shape	Scale	P-Value
ARCH Lag[3]	0.7476	0.500	2.000	0.3872
ARCH Lag[5]	2.0992	1.440	1.667	0.4497
ARCH Lag[7]	2.7705	2.315	1.543	0.5579

Nyblom stability test

Joint Statistic: 82.1798

Individual Statistics:

mu 4.0673

omega 11.2657

alpha1 0.4426

beta1 0.4927

Asymptotic Critical Values (10% 5% 1%)

Joint Statistic: 1.07 1.24 1.6

Individual Statistic: 0.35 0.47 0.75

Sign Bias Test

	t-value	prob	sig
Sign Bias	1.16053	0.2460	
Negative Sign Bias	1.24643	0.2128	
Positive Sign Bias	0.04017	0.9680	
Joint Effect	1.69790	0.6374	

Adjusted Pearson Goodness-of-Fit Test:

	group	statistic	p-value(g-1)
1	20	251.2	1.792e-42
2	30	302.2	2.949e-47
3	40	319.2	1.105e-45
4	50	336.6	1.567e-44

Elapsed time : 0.04816198

GARCH Model Performance

RMSE: 0.00568

MAE : 0.004518

The ARIMAX model that was used by the `auto.arima()`, `ARIMA(5,1,4)`, with the three macro surprises included as the predictors. As mentioned before, daily SPY volatility is dependent on its own past differences and surprises, as well as the same day CPI, unemployment, and Fed fund rate surprises. The estimated coefficients on the macro variables were very small: `cpi_surprise` (CPI surprises) -0.0038, `unemp_surprise` (unemployment surprises) 0, and `ffr_surprise` (Fed funds rate surprises) -0.0037. These coefficients were not statistically significant due to their standard errors being larger than the point estimates. This result suggests that once we account for how volatility depends on its own past, the same-day CPI, unemployment, and Fed funds rate surprises do not have a big effect on daily SPY volatility. On the test data, the ARIMAX model had an RMSE of about 0.00608 and MAE of about 0.00403. Since these metrics have a small value, this means that the model is doing a good job overall because the numbers outputted by the model are approximately the same size as the typical level of daily volatility in our data. When we look at the plot of predictions versus actual volatility, the ARIMAX model reveals the general ups and downs and the periods where volatility stays high or low, but it still fails to fully capture the most extreme spikes.

On the other hand, the GARCH(1,1) model gave parameter estimates that look normal for financial data. The estimated mean of volatility was about 0.0069. For the variance part (alpha), it was about 0.11 – this controls

how much yesterday’s squared shock matters. Also, beta, which controls how much yesterday’s variance matters, was about 0.88. The sum of alpha and beta was roughly around 0.99, which is close to 1. This tells us that volatility is highly persistent, and when volatility becomes high, it will stay high for a while before slowly coming back down. On the test set, the GARCH model had an RMSE of about 0.00567 and an MAE of about 0.00445. Comparing the GARCH model to the ARIMAX model, GARCH has a slightly lower RMSE but a slightly higher MAE. These results reveal that GARCH is a bit better at capturing large volatility spikes, which would affect the RMSE. On the other hand, ARIMAX does a better job at capturing smaller day-to-day movements, which MAE focuses on.

Overall, when looking at the comparison for both models, ARIMAX and GARCH, we can see that they capture the main patterns in SPY volatility, and their performance on the new data were pretty similar. GARCH(1,1) has a slightly better performance in RMSE because how well the model can capture volatility clustering and extreme moves more efficiently. ARIMAX has the better end when it comes to MAE because on normal days, its predictions are slightly closer on average. With this information and the fact that the macro surprise coefficients in the ARIMAX model are small and not statistically significant, we can confidently say that most of the short-term predictability in SPY volatility comes from its own past behavior, rather from monthly macroeconomic surprises. Although macro announcements still matter, the day-to-day volatility we see in the market is driven more by internal market dynamics than by any single CPI, unemployment, or Fed release.

5. Conclusion –

In this project, we examined how month-to-month changes in major U.S. macroeconomic indicators—CPI, the unemployment rate, and the Federal Funds Rate—affect short-term volatility in the SPY ETF. Using a combination of exploratory data analysis, regression modeling, and time-series methods, we identified several important patterns about how financial markets respond to new economic information.

Our regression models show that while macroeconomic surprises do influence volatility, the effects are generally small. CPI and unemployment surprises rarely produce statistically significant changes in SPY volatility, which is consistent with their highly stable and predictable month-to-month behavior (as shown in our EDA). In contrast, surprises in the Federal Funds Rate—an indicator known for producing sudden policy shocks—show a modest but statistically significant relationship with short-term volatility. This aligns with economic intuition: markets react more strongly when the Federal Reserve adjusts interest rates.

Our ARIMA time-series model provided a complementary perspective by forecasting future volatility using only SPY’s historical patterns. Diagnostics showed that the ARIMA(2,1,5) model captured most of the autocorrelation structure, and the forecast indicated a stable return toward average volatility levels. This suggests that much of the day-to-day volatility is driven by internal market dynamics rather than macroeconomic announcements. The comparison between regression and ARIMA models highlights a key finding: past volatility explains far more of today’s volatility than newly released macroeconomic data. This aligns with the broader literature on volatility clustering.

There are several limitations worth noting. First, we only modeled short-term volatility around release dates, so longer reaction windows may reveal different effects. Second, macroeconomic data—especially CPI and unemployment—are slow-moving by nature, which reduces their explanatory power in predicting rapid shifts in volatility. Finally, our models do not incorporate other major market events (earnings cycles, geopolitical events, market sentiment) that may overshadow macro releases.

Future extensions for this project could include: Expanding the event window to examine multi-day volatility spillovers, Using a GARCH-X model to directly incorporate macro surprises into the volatility equation, Separating high-volatility and low-volatility regimes using a Markov-switching model, Applying machine learning models—such as random forests—to capture nonlinear relationships

Overall, our findings suggest that while macroeconomic announcements matter, their short-term impact on volatility is modest compared to the effect of market-driven volatility dynamics.

6. Author Contribution Statement

Dhruv Nasit, Felicia Vijayarangam, Joseph Doan, and Eric Feng jointly conceived the project and defined the research question.

Dhruv Nasit developed the computational workflow, performed all data merging and cleaning for the macroeconomic and SPY datasets, carried out the exploratory data analysis, implemented the event-study volatility calculations, built the regression models, and generated all associated figures.

Felicia Vijayarangam performed the initial preprocessing of the macroeconomic datasets described in the Data Description section, contributed to the EDA setup prior to modeling, and wrote portions of the Methodology and Data Analysis sections related to the regression framework.

Joseph Doan drafted the Introduction and the Time-Series Methodology section, contributed to the written interpretation of both the ARIMA and GARCH models, and assisted in organizing the structure and narrative of the full report.

Eric Feng developed the ARIMA and GARCH time-series models, produced the accompanying diagnostics and interpretations, and contributed feedback on the statistical validity and clarity of the modeling results.

All authors discussed the findings, contributed to the interpretation of the results, and helped revise the final manuscript.

7. Code Appendix

```
library(tidyverse)
library(lubridate)
library(gridExtra)

# -----
# 1. Load datasets
# -----
cpi <- read.csv("cpi_cleaned.csv")
fed <- read.csv("fed_rate_cleaned.csv")
unemp <- read.csv("unemployment_cleaned.csv")
spy <- read.csv("SPY_data.csv")

head(cpi, 2)

head(spy, 2)
spy <- spy %>%
  arrange(X) %>%
  mutate(ret = log(SPY.Adjusted / lag(SPY.Adjusted)),
         spy_vol = abs(ret))
colnames(spy)
cpi %>%
  filter(is.na(Actual) | is.na(Previous)) %>%
  select(Date, Actual, Previous)
fed %>%
  filter(is.na(Actual) | is.na(Previous)) %>%
  select(Date, Actual, Previous)
unemp %>%
  filter(is.na(Actual) | is.na(Previous)) %>%
  select(Date, Actual, Previous)
spy %>%
  filter(if_any(everything(), is.na)) %>%
  select(X, everything())
spy <- spy %>%
  drop_na()
fed <- fed %>%
  drop_na()
unemp <- unemp %>%
  drop_na()
cpi <- cpi %>%
  drop_na()
#Convert dates + compute Diff
process_macro <- function(df, name) {
  df %>%
    mutate(
      Date = as.Date(Date, format = "%b %d, %Y"),
```

```

    Diff = Actual - Previous,
    Indicator = name
  )
}

cpi <- process_macro(cpi, "CPI")
fed <- process_macro(fed, "Fed")
unemp <- process_macro(unemp, "Unemployment")
macro_all <- bind_rows(cpi, fed, unemp)

head(macro_all, 1)

#Summary statistics
summary_stats <- macro_all %>%
  group_by(Indicator) %>%
  summarise(
    n = n(),
    mean_actual = mean(Actual, na.rm = TRUE),
    sd_actual = sd(Actual, na.rm = TRUE),
    mean_diff = mean(Diff, na.rm = TRUE),
    sd_diff = sd(Diff, na.rm = TRUE),
    min_diff = min(Diff, na.rm = TRUE),
    max_diff = max(Diff, na.rm = TRUE)
  )

print(summary_stats)

#Time-series plots of Diff
p_cpi <- ggplot(cpi, aes(Date, Diff)) +
  geom_line(color = "orange") +
  labs(title = "CPI Month-to-Month Change (Diff)",
    y = "Actual - Previous", x = "")

grid.arrange(p_cpi, ncol = 1)

p_fed <- ggplot(fed, aes(Date, Diff)) +
  geom_line(color = "steelblue") +
  labs(title = "Fed Funds Rate Change (Diff)",
    y = "Actual - Previous", x = "")

grid.arrange(p_fed, ncol = 1)

p_unemp <- ggplot(unemp, aes(Date, Diff)) +
  geom_line(color = "darkgreen") +
  labs(title = "Unemployment Rate Change (Diff)",
    y = "Actual - Previous", x = "")

grid.arrange(p_unemp, ncol = 1)

#Histogram for Diff by indicator
ggplot(macro_all, aes(Diff, fill = Indicator)) +
  geom_histogram(alpha = 0.6, bins = 40) +
  facet_wrap(~Indicator, scales = "free") +
  labs(title = "Distribution of Month-to-Month Changes (Diff)",
    x = "Diff", y = "Count") +
  theme_minimal()

#Boxplots to compare volatility of Diff
ggplot(macro_all, aes(Indicator, Diff, fill = Indicator)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Comparison of Diff Across Indicators",
    y = "Month-to-Month Change (Diff)", x = "") +
  theme_minimal()

```

```

#Overlay all 3 indicators' Diff in one plot
ggplot(macro_all, aes(Date, Diff, color = Indicator)) +
  geom_line(alpha = 0.8) +
  labs(title = "Overlay of Diff for CPI, Fed, Unemployment",
        y = "Actual - Previous", x = "") +
  theme_minimal()
#Scatter: Actual vs Previous
ggplot(macro_all, aes(Previous, Actual, color = Indicator)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Actual vs Previous by Indicator",
        x = "Previous", y = "Actual") +
  theme_minimal()
#2008-2015
cpi <- cpi %>%
  mutate(Date = as.Date(Date, format = "%b %d, %Y")) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))

unemp <- unemp %>%
  mutate(Date = as.Date(Date, format = "%b %d, %Y")) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))

fed <- fed %>%
  mutate(Date = as.Date(Date, format = "%b %d, %Y")) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))

spy <- spy %>%
  rename(Date = X) %>%
  mutate(Date = as.Date(Date)) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))
#Dummy=0 prior to the monthly release
#Dummy=1 on and after the release date but only within that month.
#Diff is set to the release difference for the active days, otherwise 0.

expand_month_limited <- function(daily_dates, releases_df, diff, prefix) {
  # Keep Release date and the difference
  rel <- releases_df %>%
    select(Date, !!sym(diff)) %>%
    mutate(
      month_index = floor_date(Date, "month"),
      release_date = Date,
      release_diff = !!sym(diff)
    ) %>%
    select(month_index, release_date, release_diff)

  # Daily panel with attached monthly release info
  df <- tibble(Date = daily_dates) %>%
    mutate(month_index = floor_date(Date, "month")) %>%
    left_join(rel, by = "month_index") %>%
    mutate(
      # Dummy=1 only on and after the monthly release date within same month
      dummy = ifelse(
        !is.na(release_date) &
        Date >= release_date &
        Date <= (month_index + months(1) - days(1)),
        1, 0
      ),
      # active difference only when dummy==1

```

```

    diff_active = ifelse(dummy == 1, release_diff, 0)
  ) %>%
  select(
    Date,
    !!paste0(prefix, "_diff") := diff_active,
    !!paste0(prefix, "_day") := dummy
  )

  return(df)
}
#Apply
daily_dates <- spy$Date

cpi_expanded <- expand_month_limited(
  daily_dates = daily_dates,
  releases_df = cpi,
  diff = "Diff",
  prefix = "cpi"
)

unemp_expanded <- expand_month_limited(
  daily_dates = daily_dates,
  releases_df = unemp,
  diff = "Diff",
  prefix = "unemp"
)

fed_expanded <- expand_month_limited(
  daily_dates = daily_dates,
  releases_df = fed,
  diff = "Diff",
  prefix = "fed"
)

#Merge into final dataset
data <- spy %>%
  left_join(cpi_expanded, by = "Date") %>%
  left_join(unemp_expanded, by = "Date") %>%
  left_join(fed_expanded, by = "Date") %>%
  mutate(
    cpi_diff = replace_na(cpi_diff, 0),
    unemp_diff = replace_na(unemp_diff, 0),
    fed_diff = replace_na(fed_diff, 0),
    cpi_day = replace_na(cpi_day, 0),
    unemp_day = replace_na(unemp_day, 0),
    fed_day = replace_na(fed_day, 0),
    all_three = ifelse(cpi_day + unemp_day + fed_day == 3, 1, 0)
  )

#Run linear regression
model1 <- lm(spy_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day + fed_diff + fed_day + all_three,
  data = data)

summary(model1)
model_summary <- summary(model1)

r2 <- model_summary$r.squared
adj_r2 <- model_summary$adj.r.squared
aic_val <- AIC(model1)

```



```

bic_val <- BIC(model1)
# ---- RMSE & MAE ----
preds <- model1$fitted.values
actual <- model1$model[, 1] # first column is the dependent variable

rmse_val <- sqrt(mean((actual - preds)^2))
mae_val <- mean(abs(actual - preds))

cat("R²: ", r2, "\n")
cat("Adjusted R²: ", adj_r2, "\n")
cat("AIC: ", aic_val, "\n")
cat("BIC: ", bic_val, "\n")
cat("RMSE: ", rmse_val, "\n")
cat("MAE: ", mae_val, "\n")

library(lmtest)
library(sandwich)
library(dplyr)

# ---- Robust SE ----
robust <- coeftest(model1, vcov = vcovHC(model1, type = "HC1"))

robust_df <- data.frame(
  term = rownames(robust),
  estimate = robust[, 1],
  std.error = robust[, 2],
  t.value = robust[, 3],
  p.value = robust[, 4]
)

# ---- Sensitivity Table for Surprise Variables ----
sens <- robust_df %>%
  filter(term %in% c("cpi_diff", "unemp_diff", "fed_diff")) %>%
  mutate(pct_change_vol = estimate * 100)

sens

# ---- Interpretation Functions ----
interpret_effect <- function(name, est, se) {
  sprintf(
    "A one-unit %s difference changes daily SPY volatility by %.2f%% (robust SE = %.4f).",
    name, est * 100, se
  )
}

interpret_day_effect <- function(term, pct) {
  event_name <- dplyr::case_when(
    term == "cpi_day" ~ "CPI release days",
    term == "unemp_day" ~ "Unemployment report days",
    term == "fed_day" ~ "Fed rate decision days",
    TRUE ~ term
  )

  sprintf(
    "On %s, daily SPY volatility is estimated to change by %.2f%%.",
    event_name, pct
  )
}

# ---- Print Surprise/Difference Effects ----

```

```

cat(
  interpret_effect(
    "CPI",
    robust_df$estimate[robust_df$term == "cpi_diff"],
    robust_df$std.error[robust_df$term == "cpi_diff"]
  ),
  "\n"
)

cat(
  interpret_effect(
    "Unemployment",
    robust_df$estimate[robust_df$term == "unemp_diff"],
    robust_df$std.error[robust_df$term == "unemp_diff"]
  ),
  "\n"
)

cat(
  interpret_effect(
    "Fed Funds Rate",
    robust_df$estimate[robust_df$term == "fed_diff"],
    robust_df$std.error[robust_df$term == "fed_diff"]
  ),
  "\n"
)
# ---- Print Event-Day Effects ----
cat(
  interpret_day_effect(
    "cpi_day",
    robust_df$estimate[robust_df$term == "cpi_day"] * 100
  ),
  "\n"
)

cat(
  interpret_day_effect(
    "unemp_day",
    robust_df$estimate[robust_df$term == "unemp_day"] * 100
  ),
  "\n"
)

cat(
  interpret_day_effect(
    "fed_day",
    robust_df$estimate[robust_df$term == "fed_day"] * 100
  ),
  "\n"
)
#Special Case: all_three
event_interpret_three <- robust_df %>%
  filter(term == "all_three") %>%
  mutate(
    pct_change = estimate * 100,
    interpretation = paste0(
      "On days when CPI, unemployment, and the Fed decision are all released, ",
      "volatility changes by ", round(pct_change, 2), "% on the event day."
    )
  )

```

```

    )
  )

event_interpret_three$interpretation
# Compute median
vol_median <- median(data$spy_vol, na.rm = TRUE)

data$high_vol <- ifelse(data$spy_vol > vol_median, 1, 0)

cat("Median of spy_vol:", vol_median, "\n\n")

cat("Label counts:\n")
print(table(data$high_vol))

cat("\nLabel proportions:\n")
print(prop.table(table(data$high_vol)))

cat("\nPreview of labels:\n")
print(head(data[, c("spy_vol", "high_vol")]))
library(caret)
library(pROC)
#Train/Test
set.seed(123)

train_index <- createDataPartition(data$spy_vol, p = 0.8, list = FALSE)

train_data <- data[train_index, ]
test_data  <- data[-train_index, ]

#Logistic Regression
log_model <- glm(
  high_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day + fed_diff + fed_day + all_three,
  data = train_data,
  family = binomial
)

summary(log_model)
model_summary <- summary(log_model)

r2 <- model_summary$r.squared
adj_r2 <- model_summary$adj.r.squared
aic_val <- AIC(log_model)
bic_val <- BIC(log_model)
# ---- RMSE & MAE ----
preds <- model1$fitted.values
actual <- model1$model[, 1] # first column is the dependent variable

rmse_val <- sqrt(mean((actual - preds)^2))
mae_val  <- mean(abs(actual - preds))

cat("R²: ", r2, "\n")
cat("Adjusted R²: ", adj_r2, "\n")
cat("AIC: ", aic_val, "\n")
cat("BIC: ", bic_val, "\n")
cat("RMSE: ", rmse_val, "\n")
cat("MAE: ", mae_val, "\n")
# Predict probabilities of high volatility
test_probs <- predict(log_model, newdata = test_data, type = "response")

```

```

# Convert probabilities → class labels using 0.5 threshold
test_pred <- ifelse(test_probs > 0.5, 1, 0)

# Convert to factor so confusionMatrix works
test_pred <- factor(test_pred, levels = c(0,1))
true_labels <- factor(test_data$high_vol, levels = c(0,1))

#Confusion Matrix + Metrics

confusionMatrix(
  data = as.factor(test_pred),
  reference = as.factor(true_labels),
  positive = "1"      # high volatility = positive class
)

library(pROC)
test_roc <- roc(
  response = true_labels,      # actual labels (0/1)
  predictor = test_probs,      # predicted probabilities
  plot = TRUE,                 # draw ROC curve
  print.auc = TRUE,            # print AUC on plot
  legacy.axes = TRUE           # optional: makes axes 1-specificity
)

# Extract numeric AUC
as.numeric(test_roc$auc)

library(forecast)
library(rugarch)
xreg_numeric <- data %>%
  select(cpi_diff, unemp_diff, fed_diff)

#Train / Test Split
set.seed(380) # for reproducibility of any internal routines

n <- nrow(data)
split <- floor(0.7 * n)

train_vol <- data$spy_vol[1:split]
test_vol <- data$spy_vol[(split + 1):n]

train_x <- as.matrix(xreg_numeric[1:split, ])
test_x <- as.matrix(xreg_numeric[(split + 1):n, ])

cat("TRAIN:", length(train_vol), "obs   TEST:", length(test_vol), "obs\n")

#ARIMAX model (volatility ~ own lags + macro surprises)

spy_vol_ts_train <- ts(train_vol, frequency = 252)

arimax_model <- auto.arima(
  spy_vol_ts_train,
  xreg = train_x,
  seasonal = FALSE,
  stepwise = TRUE,
  approximation = FALSE
)

```

```

cat("\n===== ARIMAX Model =====\n")
print(arimax_model)

# Forecast on test period
arimax_fc <- forecast(arimax_model, xreg = test_x, h = length(test_vol))$mean

arimax_rmse <- sqrt(mean((arimax_fc - test_vol)^2))
arimax_mae <- mean(abs(arimax_fc - test_vol))

cat("\nARIMAX Model Performance\n")
cat("RMSE:", round(arimax_rmse, 6), "\n")
cat("MAE :", round(arimax_mae, 6), "\n")

#Model Comparison + Coeffs
#GARCH(1,1) model on volatility (no xreg)
garch_spec <- ugarchspec(
  variance.model = list(
    model = "sGARCH",
    garchOrder = c(1, 1)
  ),
  mean.model = list(
    armaOrder = c(0, 0),
    include.mean = TRUE
  ),
  distribution.model = "norm"
)

garch_fit <- ugarchfit(spec = garch_spec, data = train_vol)

cat("\n===== GARCH(1,1) Fit =====\n")
show(garch_fit)

# Forecast for test period
garch_fc <- ugarchforecast(
  garch_fit,
  n.ahead = length(test_vol)
)

# conditional mean of volatility
garch_fc_values <- as.numeric(fitted(garch_fc))
garch_rmse <- sqrt(mean((garch_fc_values - test_vol)^2))
garch_mae <- mean(abs(garch_fc_values - test_vol))

cat("\nGARCH Model Performance\n")
cat("RMSE:", round(garch_rmse, 6), "\n")
cat("MAE :", round(garch_mae, 6), "\n")

Model_Comparison <- data.frame(
  Model = c("ARIMAX", "GARCH(1,1)"),
  RMSE = c(arimax_rmse, garch_rmse),
  MAE = c(arimax_mae, garch_mae)
)

```